

Rapid isolation of mitochondrial DNA. Mitochondrial DNA from *Drosophila serrata*

P. Pissios and Z. G. Scouras (1971)

Department of Genetics, Development and Molecular Biology, School of Biology, Faculty of Sciences, Aristotelian University of Thessaloniki, Thessaloniki 54006 (Greece)

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Abstract. A simple and rapid method for isolation of high quality mitochondrial DNA (mtDNA) is presented in this report. Using this method, isolation and restriction site maps for 10 enzymes of the mtDNA of *Drosophila serrata* were established.

Key words. Mitochondrial DNA; isolation; *Drosophila serrata*.

Analysis of mtDNA has proved to be a powerful tool in many fields of biology. Simple and rapid methods for mtDNA isolation are needed especially in population genetics where many samples have to be examined. High quality mtDNA is essential, so that digestion enzyme analyses and Southern blot hybridizations yield unequivocal results. We describe here a rapid and efficient two-step method of mtDNA isolation: a) the purification of the mitochondria, avoiding the nuclear DNA, and b) the isolation of the mtDNA. The reagents used in this protocol are very common in every laboratory¹.

Using the above method, the mtDNA of *Drosophila serrata*, a species belonging to the *montium* subgroup of the

melanogaster species group^{2,3}, was isolated and digested with several restriction enzymes. Both restriction enzyme analyses, and hybridizations mtDNA fragments either from *D. serrata* or from *D. melanogaster* as probes, result in a detailed map for 10 restriction endonucleases.

Materials and methods

Animals. *Drosophila serrata*, strain No. 3018.1 from the University of Texas Stock Center³ and *D. melanogaster*, Canton/S (C/S) were used in the present study.

Isolation of the mtDNA. The method used to isolate the *Drosophila* mtDNA, is described below: